Human Variant Calling and Functional Annotation Using GATK and SnpEff

FASTQ → QC → Trimming → Alignment → Variant Calling → Annotation → IGV Validation

Project Goal

Develop an end-to-end variant calling workflow for identifying and interpreting genomic variants from NGS sequencing data.

Objectives

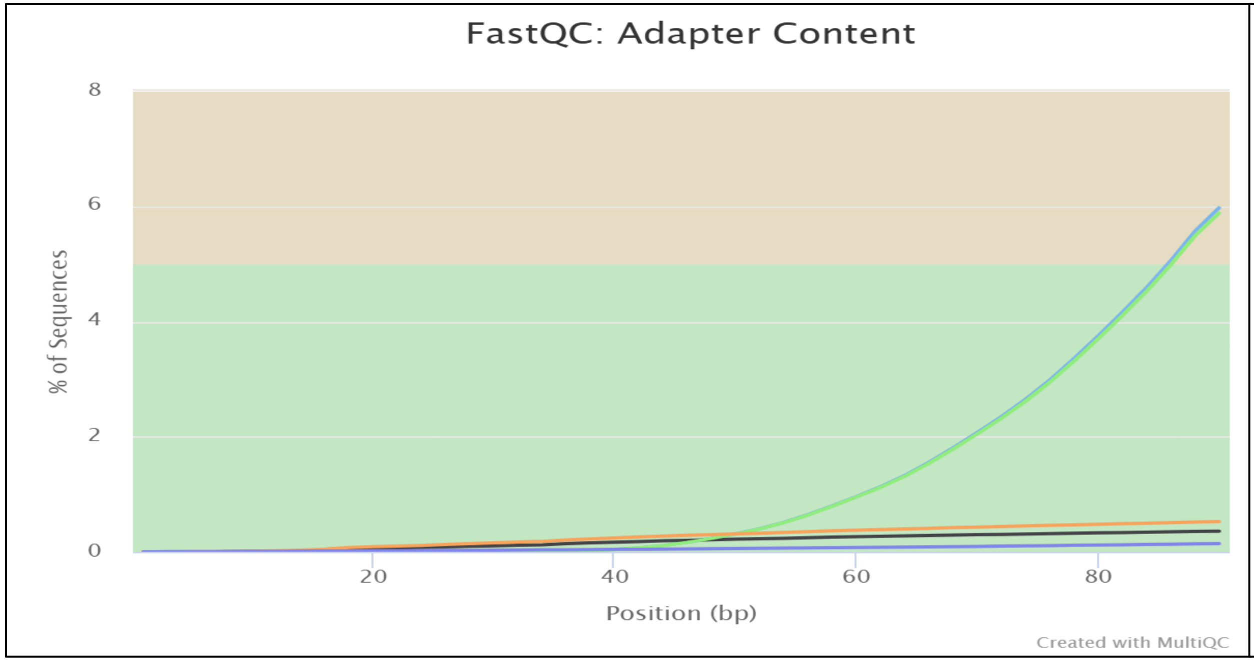
-- Perform quality assessment of raw paired-end FASTQ reads.
-- Trim low-quality bases and adapter sequences.
-- Align reads against a selected subset of GRCh38 chromosomes.
-- Process alignments and remove duplicate reads.
-- Detect genomic variants using GATK HaplotypeCaller.
-- Apply quality-based variant filtering.
-- Perform functional annotation using SnpEff.
-- Validate representative variants using IGV.

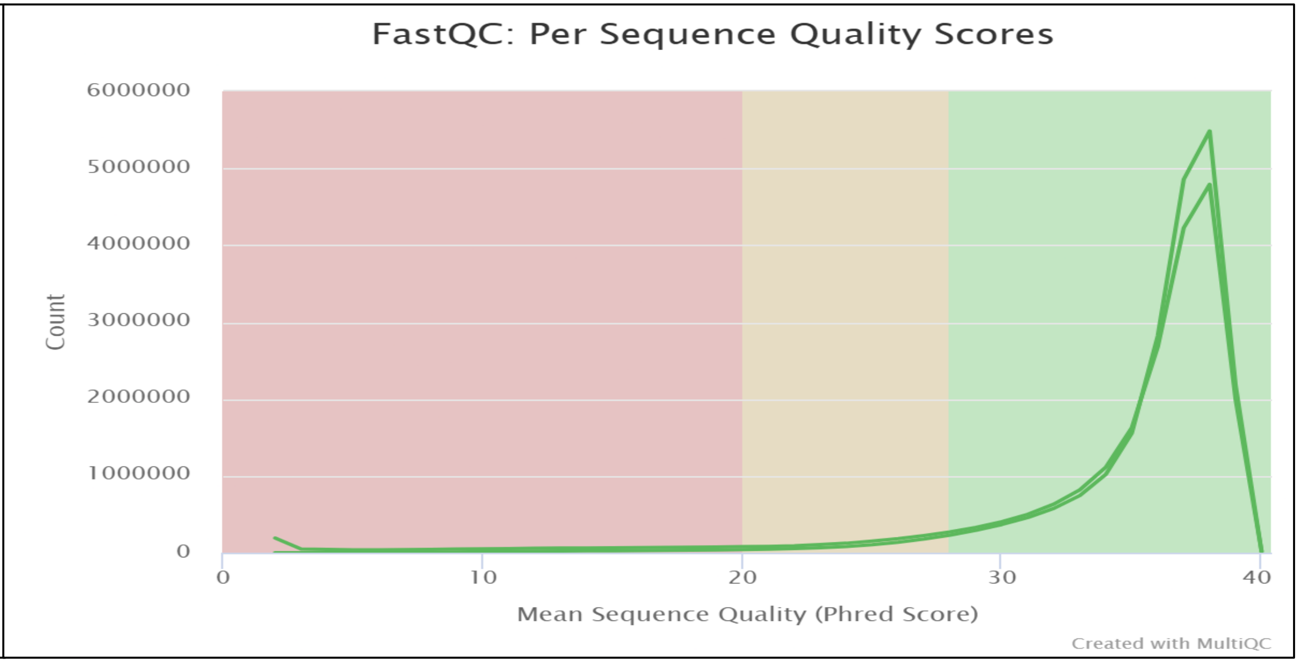
Note: A subset of the human reference genome was used for computational efficiency and local hardware limitations.

Workflow Diagram

FASTQ
↓
QC (FastQC +
MultiQC)
↓
Trimming (Fastp)
↓
Alignment (BWA)
↓
Post-processing
(SAMtools + GATK)
↓
Variant Calling
↓
Annotation
↓
IGV Validation

Raw Data QC (BEFORE trimming)





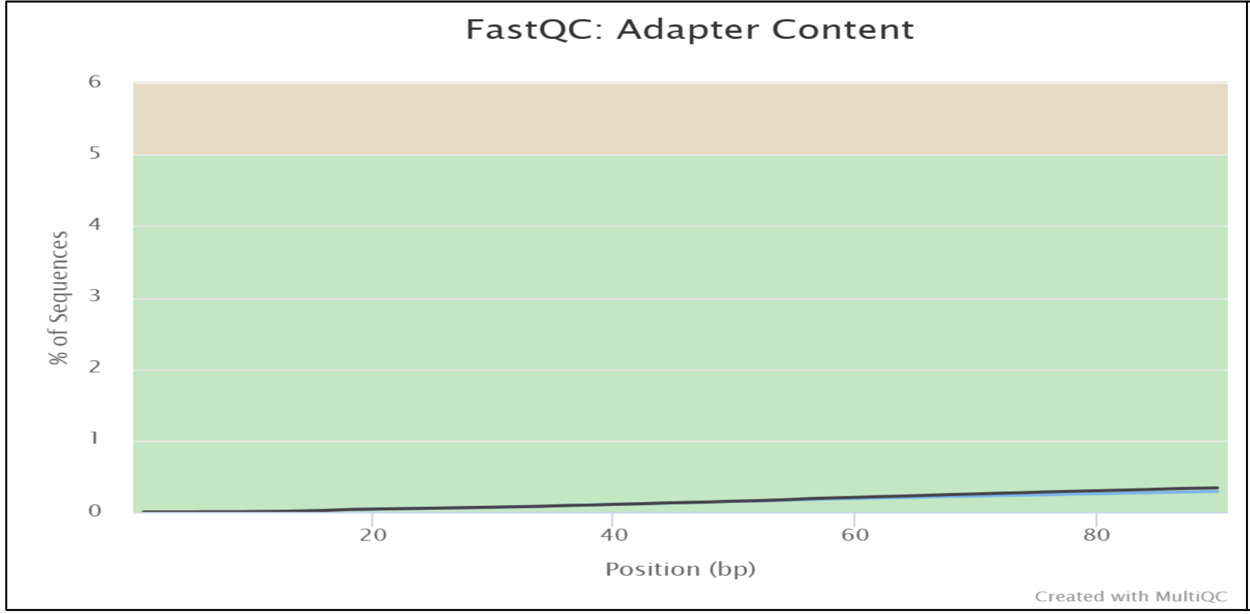
General Statistics

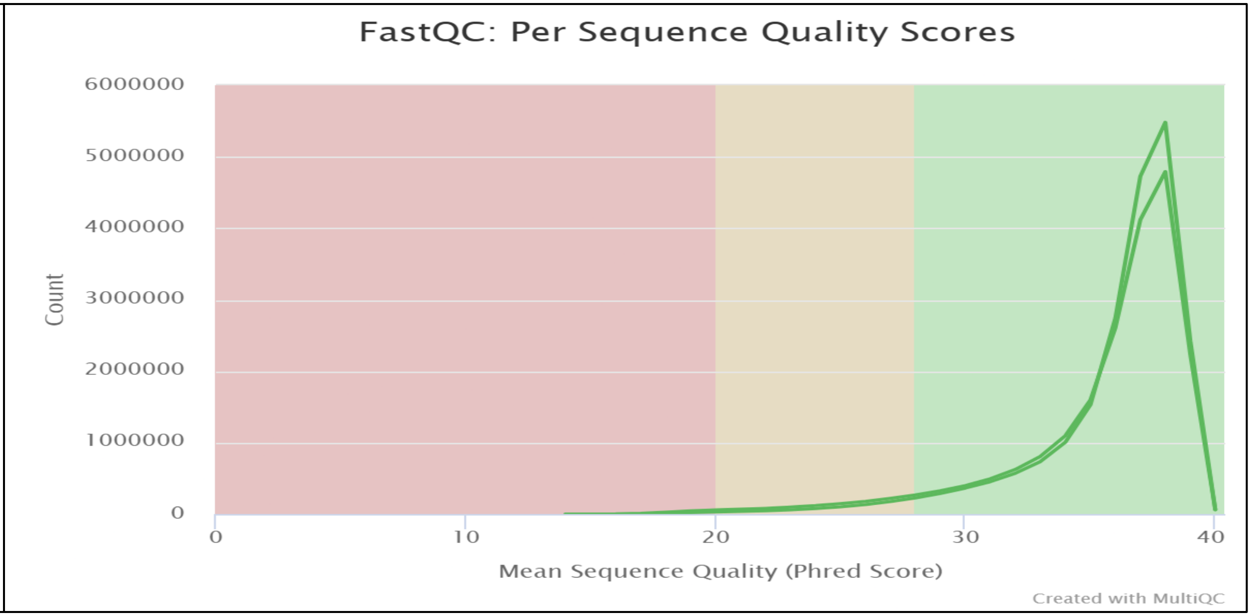
Copy table | Configure Columns | Plot | Showing 2/2 rows and 3/6 columns.

Sample Name	% Dups	% GC	M Seqs
NIST7086_CGTACTAG_L001_R1_001	30.5%	49%	21.5
NIST7086_CGTACTAG_L001_R2_001	29.4%	49%	21.5

- Adapter contamination detected.
- High overall sequencing quality.
- QC informed trimming decisions.

Trimming Results





General Statistics

Copy table | Configure Columns | Plot | Showing 2/2 rows and 3/6 columns.

Sample Name	% Dups	% GC	M Seqs
trimmed_R1	30.5%	49%	21.3
trimmed_R2	29.7%	49%	20.5

Metric	Before	After
Adapter contamination	Present	Reduced
Q30	High	Improved
Reads	21.5M	21.3M

Alignment + Duplicate Metrics

```
(variant_calling) godfather@Godfather:~/Downloads/Unix/Variant_Calling$ samtools flagstat dedup/dedup.bam
40827595 + 0 in total (QC-passed reads + QC-failed reads)
40698778 + 0 primary
0 + 0 secondary
128817 + 0 supplementary
1563137 + 0 duplicates
1563137 + 0 primary duplicates
18745680 + 0 mapped (45.91% : N/A)
18616863 + 0 primary mapped (45.74% : N/A)
40698778 + 0 paired in sequencing
20349389 + 0 read1
20349389 + 0 read2
17637824 + 0 properly paired (43.34% : N/A)
18178848 + 0 with itself and mate mapped
438015 + 0 singletons (1.08% : N/A)
263590 + 0 with mate mapped to a different chr
98160 + 0 with mate mapped to a different chr (mapQ>=5)
```

Total Reads: ~40.8 M
Mapped Reads: ~18.7 M (45.9%)
Properly Paired: ~17.6 M (43.3%)
Duplicates: ~1.56 M
Singletons: ~438 K

- Reads were aligned to hg38 subset reference.
- ~46% mapping rate observed.
- Duplicate reads identified and marked.
- Final BAM used for variant discovery.

```
## METRICS CLASS        picard.sam.DuplicationMetrics
LIBRARY UNPAIRED_READS_EXAMINED READ_PAIRS_EXAMINED     SECONDARY_OR_SUPPLEMENTARY_RDS  UNMAPPED_READS  UNPAIRED_READ_DUPLICATES        READ_PAIR_DUPLICATES
READ_PAIR_OPTICAL_DUPLICATES    PERCENT_DUPLICATION     ESTIMATED_LIBRARY_SIZE
Unknown Library 438015  9089424 128817  22081915        162733  700202  579     0.083964        55966703

## HISTOGRAM    java.lang.Double
BIN     CoverageMult    all_sets        optical_sets    non_optical_sets
1.0     1.000059        7869415 0       7869933
2.0     1.850203        434243  577     433770
3.0     2.572906        54607   1       54570
4.0     3.187272        15009   0       15004
5.0     3.709541        6115    0       6112
6.0     4.15352 3109    0       3111
7.0     4.530943        1815    0       1813
8.0     4.851789        1159    0       1159
9.0     5.124538        821     0       821
10.0    5.3564  552     0       552
11.0    5.553505        427     0       427
12.0    5.721063        300     0       300
13.0    5.863503        263     0       264
14.0    5.984591        202     0       201
15.0    6.087527        172     0       172
16.0    6.175032        128     0       128
17.0    6.24942 113     0       114
```

Read Pairs Examined: ~9.09 M
Read Pair Duplicates: ~700 K
Percent Duplication: ~8.4%
Estimated Library Size: ~16.3 M

- Duplicate reads were identified and marked.
- Duplication rate remained low (~8%).
- Data quality acceptable for downstream variant calling.

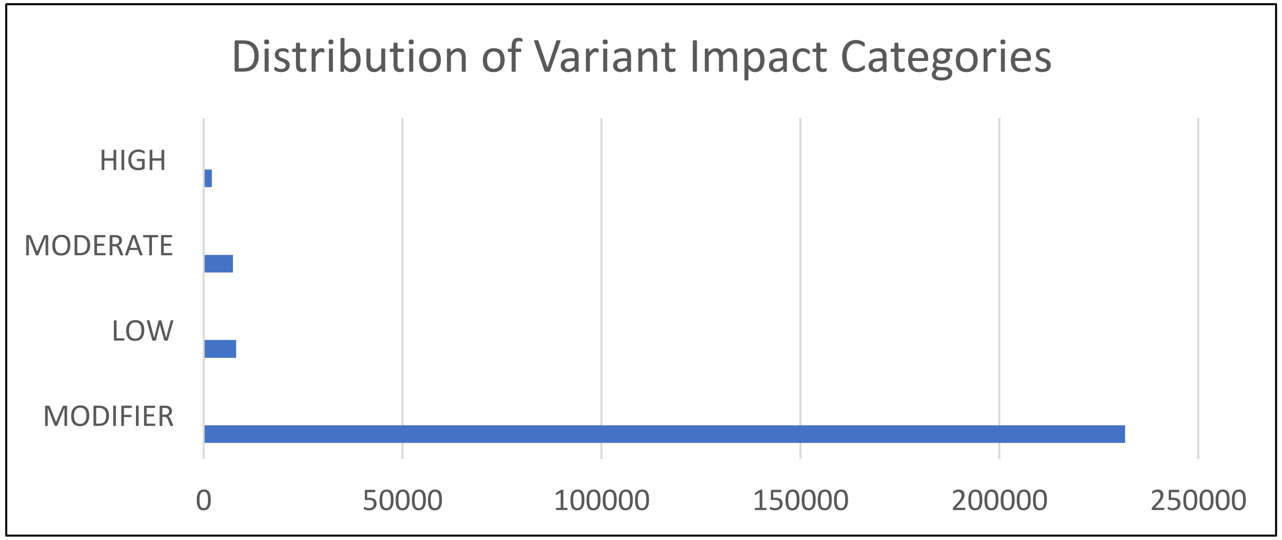
Variant Calling

```
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt Vs. Ref base qualities">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads may have been filtered">
##INFO=<ID=ExcessHet,Number=1,Type=Float,Description="Phred-scaled p-value for exact test of excess heterozygosity">
##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's exact test to detect strand bias">
##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient as estimated from the genotype likelihoods per-sample when compared again
st the Hardy-Weinberg expectation">
##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation (MLE) for the allele counts (not necessarily the same as the AC), for eac
h ALT allele, in the same order as listed">
##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation (MLE) for the allele frequency (not necessarily the same as the AF), for ea
ch ALT allele, in the same order as listed">
##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum test of Alt vs. Ref read mapping qualities">
##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank sum test of Alt vs. Ref read position bias">
##INFO=<ID=SOR,Number=1,Type=Float,Description="Symmetric Odds Ratio of 2x2 contingency table to detect strand bias">
##contig=<ID=chr1,length=248956422>
##contig=<ID=chr2,length=242193529>
##contig=<ID=chrX,length=156040895>
##source=HaplotypeCaller
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  NA12878
chr1    11161   .       A       C       35.48   .       AC=2;AF=1.00;AN=2;DP=1;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=31.00;QD=25.36;SOR=1.609   G
T:AD:DP:GQ:PL   1/1:0,1:1:3:45,3,0
chr1    11165   .       A       G       35.48   .       AC=2;AF=1.00;AN=2;DP=1;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=31.00;QD=28.73;SOR=1.609   G
T:AD:DP:GQ:PL   1/1:0,1:1:3:45,3,0
chr1    11166   .       C       G       35.48   .       AC=2;AF=1.00;AN=2;DP=1;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=31.00;QD=30.97;SOR=1.609   G
T:AD:DP:GQ:PL   1/1:0,1:1:3:45,3,0
chr1    11173   .       T       C       35.48   .       AC=2;AF=1.00;AN=2;DP=1;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=31.00;QD=27.24;SOR=1.609   G
T:AD:DP:GQ:PL   1/1:0,1:1:3:45,3,0
chr1    12198   .       G       C       462.64  .       AC=1;AF=0.500;AN=2;BaseQRankSum=4.336;DP=184;ExcessHet=0.0000;FS=40.481;MLEAC=1;MLEAF=0.500;MQ=36.16
;MQRankSum=-1.630;QD=2.58;ReadPosRankSum=-1.569;SOR=2.964       GT:AD:DP:GQ:PL  0/1:149,30:179:99:470,0,3770
chr1    12807   .       C       G       37.64   .       AC=1;AF=0.500;AN=2;BaseQRankSum=1.503;DP=90;ExcessHet=0.0000;FS=7.380;MLEAC=1;MLEAF=0.500;MQ=41.30;M
QRankSum=-4.485;QD=0.42;ReadPosRankSum=2.176;SOR=2.683  GT:AD:DP:GQ:PL  0/1:83,7:90:45:45,0,3223
chr1    12855   .       C       G       39.64   .       AC=1;AF=0.500;AN=2;BaseQRankSum=1.062;DP=23;ExcessHet=0.0000;FS=2.682;MLEAC=1;MLEAF=0.500;MQ=35.06;M
QRankSum=-3.189;QD=1.72;ReadPosRankSum=-0.610;SOR=1.609 GT:AD:DP:GQ:PL  0/1:19,4:23:47:47,0,601
chr1    13072   .       G       A       2999.06 .       AC=2;AF=1.00;AN=2;DP=94;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=50.19;QD=31.90;SOR=0.780   G
T:AD:DP:GQ:PL   1/1:0,94:94:99:3013,282,0
chr1    13327   .       G       C       365.06  .       AC=2;AF=1.00;AN=2;DP=14;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=27.46;QD=30.42;SOR=1.022   G
```

- **Variants Generated: 447,059**
- **Filtering Applied:**
- **✓ QD < 2**
- **✓ FS > 60**
- **✓ MQ < 40**

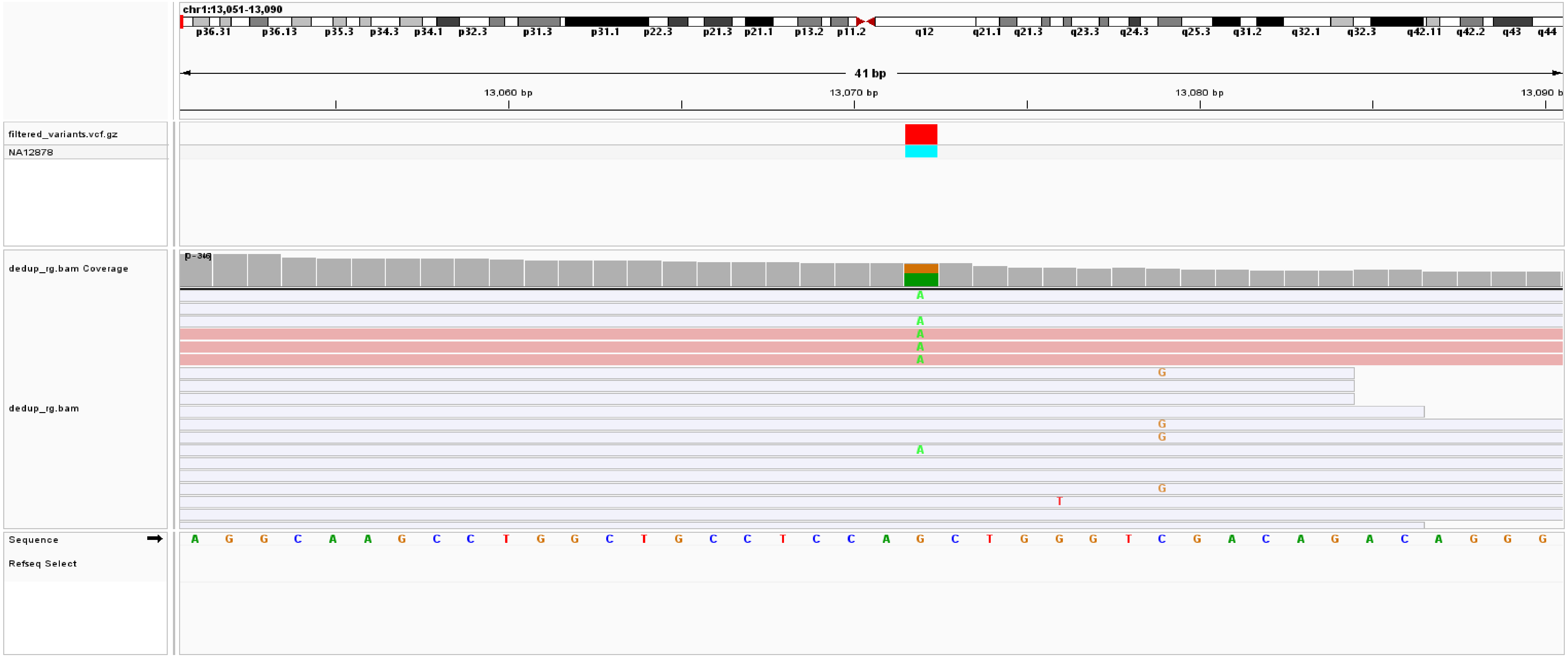
Annotation + Interpretation

```
(variant_calling) godfather@Godfather:~/Downloads/Unix/Variant_Calling$ grep -oP 'ANN=[^|]+\|\K[^|]+' \
annotation/annotated.vcf \
| sort | uniq -c | sort -nr | head
  76771 intron_variant
  48733 intergenic_region
  42514 upstream_gene_variant
  38946 downstream_gene_variant
  11965 non_coding_transcript_exon_variant
  11517 3_prime_UTR_variant
   6870 missense_variant
   6726 synonymous_variant
   1234 frameshift_variant
   1188 5_prime_UTR_variant
(variant_calling) godfather@Godfather:~/Downloads/Unix/Variant_Calling$ grep -oP 'ANN=[^|]+\|[^|]+\|\K[^|]+' \
annotation/annotated.vcf \
| sort | uniq -c
   2032 HIGH
   8185 LOW
   7334 MODERATE
 231637 MODIFIER
(variant_calling) godfather@Godfather:~/Downloads/Unix/Variant_Calling$ grep -oP 'ANN=[^|]+\|[^|]+\|[^|]+\|\K[^|]+' \
annotation/annotated.vcf \
| sort | uniq -c | sort -nr | head
   3421 ANKRD20A8P
   1976 LINC01691-AC242852.1
   1407 AC097374.1
    817 AC026273.1-AL845331.3
    710 AC073464.1-SNX18P14
    671 WDPCP
    667 AGBL4
    642 LINC01743
    628 AL845331.2-CNN2P11
    623 RNU6-1320P
(variant_calling) godfather@Godfather:~/Downloads/Unix/Variant_Calling$
```



Top effects:
Intron
Intergenic
Missense
Frameshift

IGV Validation



Visual validation of sequencing reads supporting detected variants

Conclusion

Key Outcomes

- Successfully built an end-to-end variant calling pipeline from raw sequencing reads to biological interpretation
- Performed quality assessment and preprocessing of paired-end FASTQ files using FastQC, MultiQC, and Fastp
- Aligned reads to a subset of the GRCh38 human reference genome using BWA for computational efficiency
- Processed alignments through sorting, indexing, and duplicate marking to improve variant confidence
- Identified genomic variants using GATK HaplotypeCaller and applied quality-based filtering
- Performed functional annotation using SnpEff to classify variants by impact and genomic effect
- Verified representative variant evidence through IGV visualization

This project demonstrates practical implementation of a bioinformatics variant discovery workflow under local computational constraints.